

**NETAJI SUBHASH ENGINEERING COLLEGE**  
**Technocity, Panchpota, Garia, Kolkata 700152**  
**Department: Information Technology**

Course Name: Database Management System Lab      Course Code: PCC-CS 691

**Problem 1 [PCC-CS-691.CO1]**

Write down the required DDL and DML instructions based on the following relation given below:

PATIENT (P\_ID: String (not null), P\_NAME : String, P\_DOB: date, P\_AILMENT : String, P\_BILL: number, P\_CONTACT : String, P\_ADDR : String)

Works to be done on the above Relations :--

- i. Create all the relations and insert at least five records in each table.
- ii. Find all the existing information in the table.
- iii. Find all the patients who can avail senior citizen discount.
- iv. Find all the patients who have heart ailment and are from kolkata.
- v. Find id, name, bill of all the patients who have a bill of more than Rs. 20000/- and are from outside kolkata.
- vi. Find all the patients who were born between 1980 and 1990.
- vii. Find all the patients who have a contact person in kolkata.

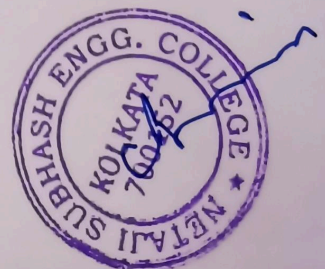
**Problem 2 [PCC- CS-691.CO1]**

Write down the required DDL and DML instructions based on the following relations given below:

SUPPLIER (SID: String (Primary key), SNAME: String, CITY: String);  
ITEM (ITEM\_ID: String (Primary key), INAME: String, COLOR: String, WEIGHT: Number);  
SUPPLY (SID: String (Primary key), ITEM\_ID: String (Primary key), QUANTITY: Number);

Works to be done on the above Relations: --

- i. Create all the relations and insert at least five records in each table.
- ii. Find all the existing suppliers.
- iii. Find all the items which are sold by at least one supplier.
- iv. Find all the suppliers who live in Delhi.
- v. Find all the suppliers who do not live in Calcutta.
- vi. Find all the suppliers who do not sell any item.
- vii. Find all the suppliers who sell at least one item.
- viii. Find all the suppliers who sell the item number 'ITEM 1'.
- ix. Find all the suppliers who sell the item number 'ITEM 1' and 'ITEM 2'.
- x. Find all the suppliers who sell either the item number 'ITEM 1' or 'ITEM 2'.
- xi. Find all the suppliers who sell the item of color RED.
- xii. Find all the suppliers who sell the item of color RED with quantity between 200 and 500.
- xiii. Find all the items which are sold by at least two suppliers.
- xiv. Find all the suppliers who sell all the items.
- xv. Find the suppliers who sell the highest quantity of one item.



### Problem 3 [PCC- CS-691.CO2]

#### Problem statement:

Data base to manage the insurance of a vehicle. The relations with their respective attributes are shown below:

Person (driver id, name, address)

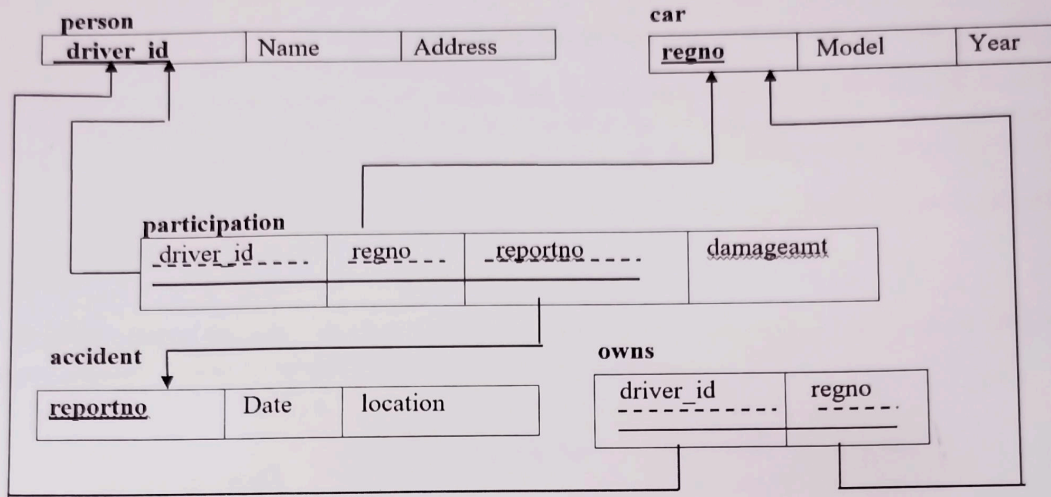
Car (regno, model, year)

Accident (reportno, date, location)

Owns (driverid, regno)

Participated (driverid, regno, reportno, damageamount)

#### Schema description:



- ▶ The table person contains the attributes namely driver\_id, name & address where the primary key is driver\_id.
- ▶ The table car contains the attributes such as model, year & register number where the primary key is register number.
- ▶ The table accident contains the attributes such as report number, date & location where the primary key is report number.
- ▶ The table owns contains the attributes driver id & regno, both the attributes are foreign key & they together constitute the composite key.

The table participated contains the attributes driver id, regno, report number & damage amount. Here driver id, regno, report number are foreign keys and they constitute the composite key.

#### Sql query:

- Query to update the damage amount for the car with a specific register number in the accident with report number between 1 & 200.
- Query to find the total number of people who owned the cars that were involved in accidents in 2002.
- Query to find the number of accidents in which cars belonging to a specific model were involved.
- Query to check if a person with a specific driver id has met with an accident in 2003.
- Query to display name of a person & the car he/she owns.



#### Problem 4 [PCC- CS-691.CO2]

##### Problem statement:

Consider the following relations for an order processing database application in a company.

**Customer** (custNo: int, cname: string, city: string)

**Order** (orderNo: int, odate: date, cust No: int, ord-amt: int)

**Order-item** (orderNo: int, item No: int, qty: int)

**Item** (itemNo: int, unit price: int)

**Shipment** (orderNo: int, warehouse No: int, ship-date: date)

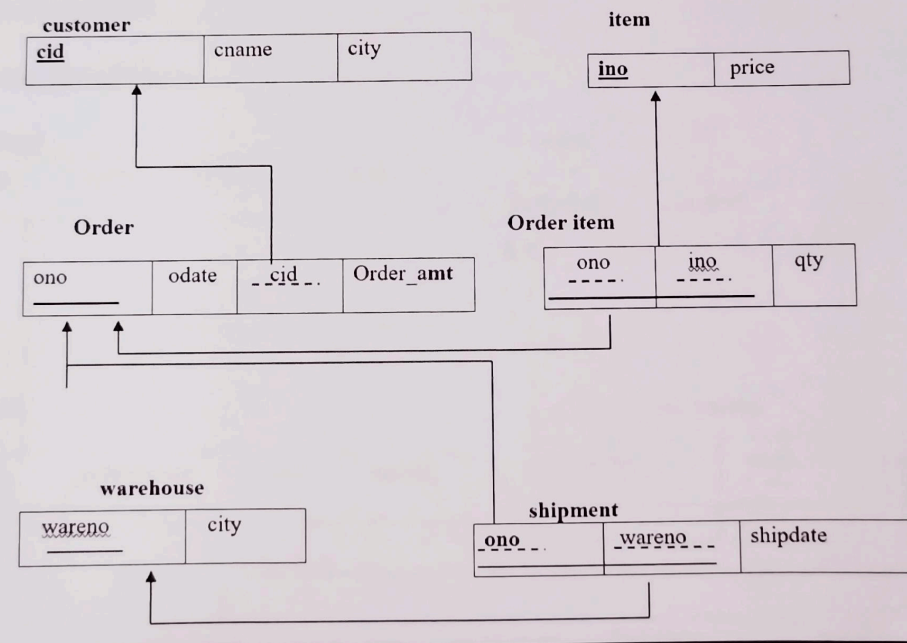
**Warehouse** (warehouseNo: int, city: string)

- Create the above tables by properly specifying the primary keys and the foreign keys.
- Enter at least five tuples for each relation.
- Produce a listing: **custname, no\_of\_orders, avg\_order\_amt**, where the middle column is the total number of orders by the customer and the last column is the average order amount for that customer.
- List the **orderNo** for the orders that were shipped from all the warehouses that the company has in a specific city.
- Demonstrate how you delete **itemNo** 10 from the **item** table and make that field *null* in the **order-item** table.

##### Schema description

The following schema describes an *order processing* database of a company. It describes the following relations: *customer, order, order-item, item, shipment and warehouse*.

##### Schema description:



### Sql queries

- ➔ Produce a listing: custname, no. of orders, avg\_order\_amt, where the middle column is the total number of orders by the customer and the last column is the average order amount for that customer.
- ➔ List the orderNo for the orders that were shipped from all the warehouses that the company has in a specific city.
- ➔ We would like to know the order numbers of all orders placed by customers who belong to a particular city.
- ➔ We would like to know the details of all orders that the company has received.

**Problem 5** [PCC- CS-691.CO3], [PCC- CS-691.CO4], [PCC- CS-691.CO5]

### Problem statement:

Consider the following database of student enrollement in courses and books adopted for each course

**Student** (regno :string , name : string , major : string , bdate : int)

**Course** (courseNo : int , cname : string , dept : string)

**Enroll** ( regno : string , courseNo: int , sem : int , marks : int )

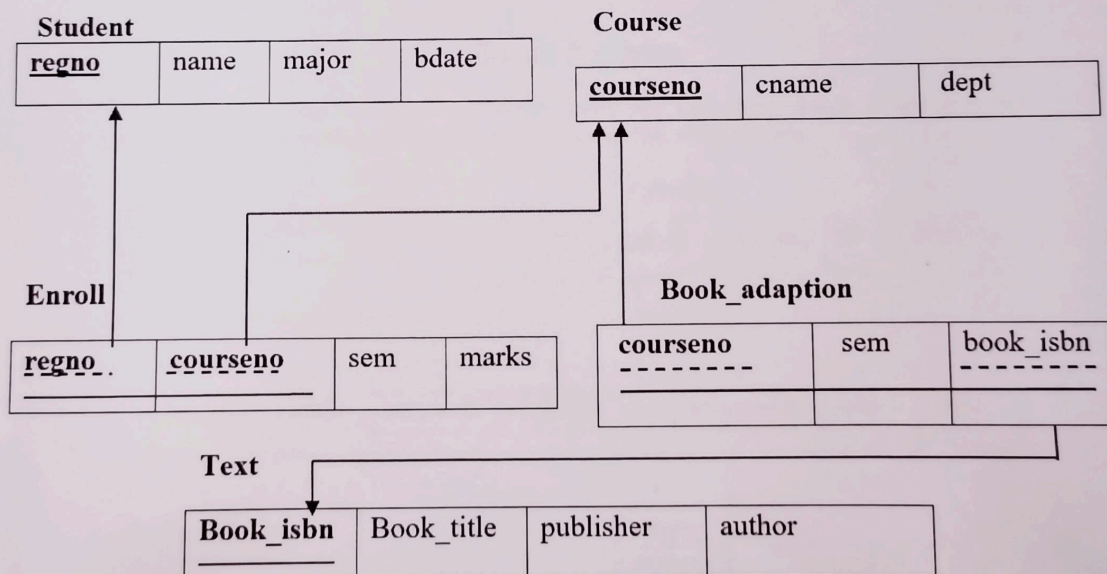
**Book\_adaption** ( courseNo: int , sem : int , book\_isbn :int)

**Text** ( book\_isbn : int , book-title : string , publisher : string , author : string).

- 1> Create the above tables by properly specifying the primary keys and the foreign key.
- 2> Enter atleast five tuples for each relation.
- 3> Solve all SQL queries.
- 4> Generation of suitable reports.

### Schema description:

### Schema description:





### SQL Queries:

- ➡ Demonstrate how you add a new text book to the database and make this book adopted by some department
- ➡ Produce a list of textbooks in the alphabetic order for courses offered by the 'cs' department that use more than two books
- ➡ List any department that has all its adopted books published by a specific publisher.

### Generation of reports

- ➡ Demonstrate how you add a new text book to the database and make this book adopted by some department
- ➡ We would like to see the number of students for each course
- ➡ We would like to see name and usn of the students of a specific course

### Problem 6 [PCC- CS-691.CO3], [PCC- CS-691.CO4]

#### Problem statement:

Consider the following relations for the details maintained by a book dealer.

**Author** (author-id: int, name: string, city: string, country: string)

**Publisher** (publisher-id: int, name: string, city: string, country: string)

**Catalog** (book-id: int, title: string, author-id: int, publisher-id: int, category-id: int, year: int, price: int)

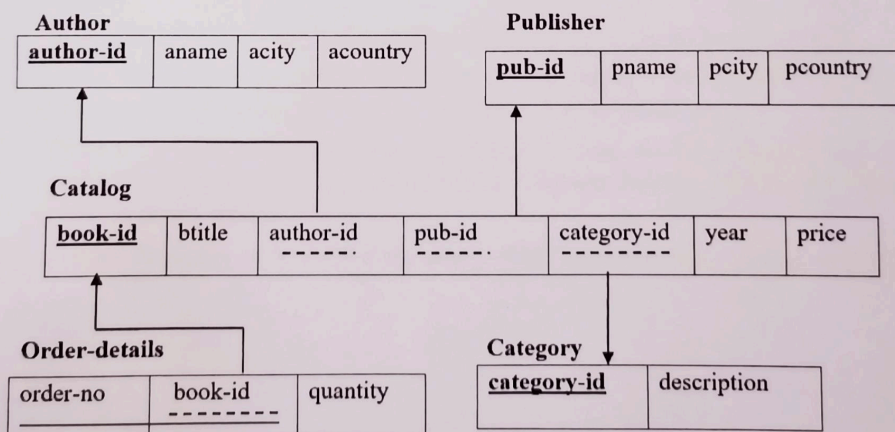
**Category** (category-id: int, description: string)

**Order-details** (order-no : int, book-id: int, quantity: int)

- a) Create the above tables by properly specifying the primary keys and the foreign keys.
- b) Enter at least five tuples for each relation.
- c) Give the details of the authors who have 2 or more books in the catalog and the price of the books is greater than the average price of the books in the catalog and the year of publication is after 2000.
- d) Find the author of the book which has maximum sales.
- e) Demonstrate how you increase the price of books published by a specific publisher by 10%.

#### Schema description

The following schema describes a database maintained by a book-dealer. It describes the following relations: *author*, *publisher*, *catalog*, *category*, *order-details*.



SQL Queries:

- This query gives details of the authors who have 2 books in the catalog and the price of the books is greater than the average price of the books in the catalog and the year of publication is after 2000.
- This query demonstrates how to retrieve the author of the book which has maximum sales.
- This query demonstrates how to increase the price of the books published by a specific publisher by 10%.

### Report

In this project we are mainly dealing with maintenance of books by a book dealer. This schema mainly includes relations like author, publisher, catalog, order\_details etc.

The queries which we have implemented includes,

Demonstration on how to retrieve the details of the authors who have 2 or more books in the catalog and their price is greater than the average price of all books in the catalog and the year of publication is after 2000.

Demonstration on how to find the author with the maximum sales.

Demonstration on how to increase the price of books published by a specific publisher.

### Problem 7 [PCC- CS-691.CO6]

Write a PL/SQL code for finding factorial of a given number

1. Write a PL/SQL code for inverting a number 5639 to 9365.
2. Write a PL/SQL code for calculating finding the sum of N numbers.
3. Write a PL/SQL code for generating a Fibonacci series.
4. Write a PL/SQL code for finds a given year is leap year or not.
5. Write a PL/SQL code for finding maximum of three numbers. (Input will be given by the use).
6. Write a PL/SQL code block to calculate the area of a circle for a value of radius varying from 6 to 10. Store the radius and corresponding values of calculated area in an empty table named Areas, Consisting of two columns Radius and Area.
7. Write a PL/SQL code block that will accept a client\_no from the user and adds the amount of Rs. 1000 to bal\_due column has a minimum balance of Rs-6000. The process is fire on client\_master.
8. Display the salary of ALLEN if the salary>5000 otherwise show proper message.